

MGT 207

Production and Operations Management



**Forecasting and Data
Analytics**

Forecasting Methods Defined

Expert Forecast (Subjective)

Qualitative Method

A method based on people's opinions.

Key Strength:

Incorporate expertise that is hard to codify.

Time-Series Forecasting

Quantitative Method

Built on the assumption that past history is the best predictor of the future.

Data Usage:

Uses historical demand data for variables.

Regression (Causal)

Statistical Method

Models relationships between dependent and independent variables.

Approach:

Assumes forecast is related to environment variables.

AI-Based (Machine Learning)

Advanced Quantitative

Based on numeric data and complex equations.

Scale:

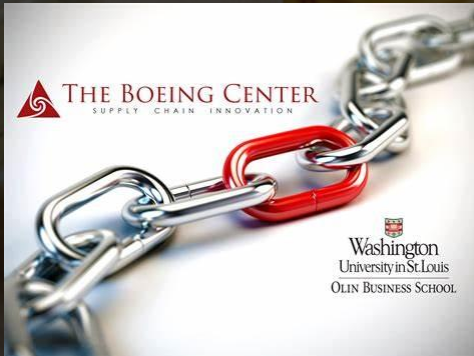
Considers large amounts of data for consistency.

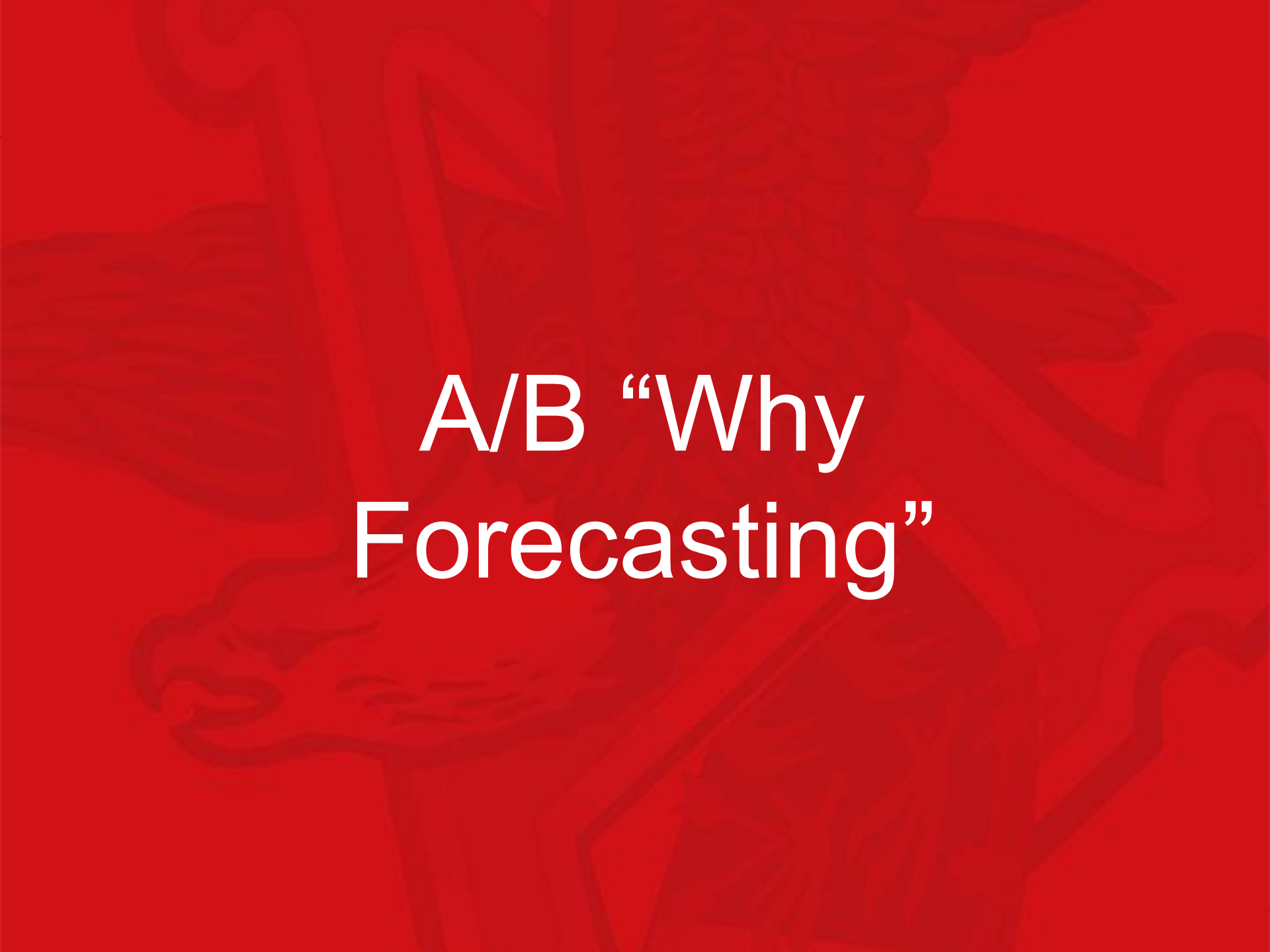


ANHEUSER-BUSCH

Example

Demand Forecasting with Machine Learning





A/B “Why Forecasting”

A/B “Why”

Forecast Accuracy directly relates to the amount of out of stocks at AB's WOD's

Total Benefit



**-4% OOS for each +1%
Accuracy**

High End Benefit



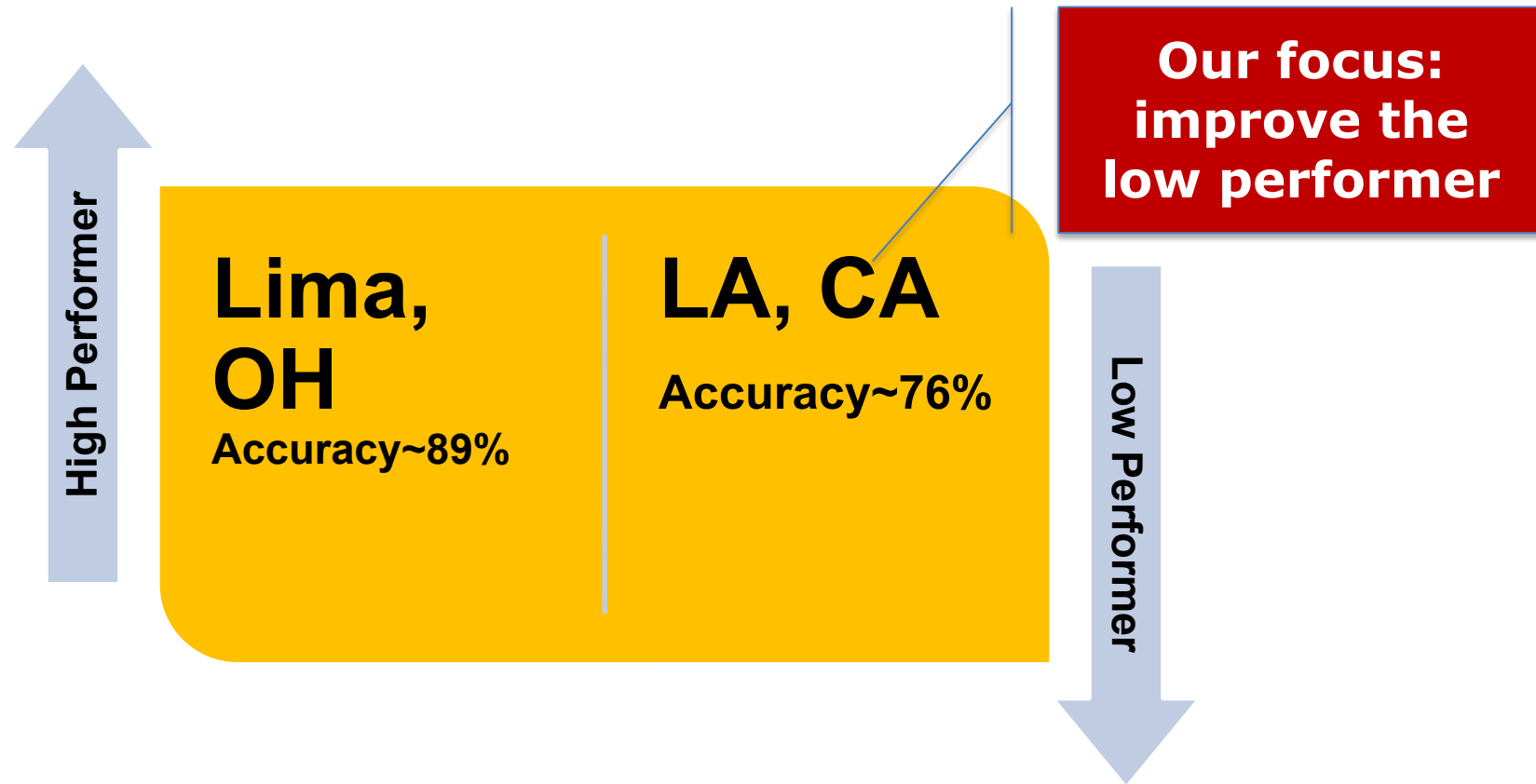
**-11% OOS for each +1%
Accuracy**



Background

Background

A/B has historically had mixed success with forecasting based on WOD location:



Time-Dependent Factors

Focusing on Factors changing over time

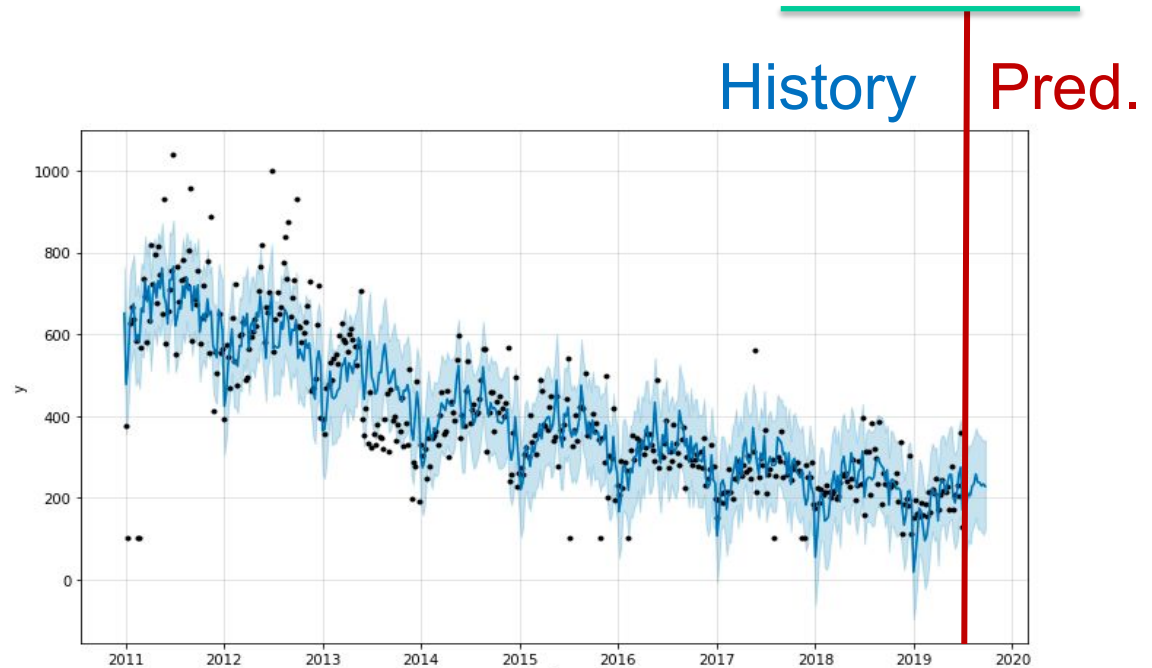
- **Sales History**
 - previous week, biweekly, monthly, 6-month, yearly
 - **Holidays**
 - Memorial Day, Labor Day, Independence Day; 2 weeks before and after
 - **Seasonality**
 - Yearly, quarterly, monthly
 - **Promotion**
 - Avg Discount
 - **Weather**
 - Avg Temp and Avg Precipitation
 - **Brand Characteristics**
 - **Frontline Price**
 - Price to retail
-

Time Series-Based Model

Use LA Data

Prediction accuracy:

~ 93%



Utilized Data:

- Past sales, seasonality, and weather

The Nuts and Bolts of **TIME SERIES FORECASTING**

Coming up next

Notations

- Demand observed in period $t-1$
 - A_{t-1}
- Forecast (made in period $t-1$) of demand for period t , i.e. forecast of coming period's demand
 - F_t

Motivational Example

Forecasting Demand for Polio Vaccines

- You have just joined a pediatric hospital as head of operations. You need to order polio vaccines monthly starting with February. You know that January demand was 130.

Month	t	Demand for Vaccine
January	1	130
February	2	
March	3	
April	4	
May	5	
June	6	
July	7	

Method 1: The Naïve Method, $F_t = A_{t-1}$

Simplest Approach to Forecasting

	Period	Demand	Forecast
Forecast of Demand in Period 2 (This forecast made after seeing Demand in Period 1)	1	130	-
	2	155	130
	3	145	155
Forecast of Demand in Period 5 (This forecast made after seeing Demand in Period 4)	4	160	145
	5	151	160
	6	143	151
	7		143

Method 2: The Simple Average,

$$F_t = (A_1 + A_2 + A_3 + \dots + A_{t-1}) / (t-1)$$

	Period	Demand	Forecast	
Forecast of Demand in Period 2 (This forecast made after seeing Demand in Period 1)	1	130	-	
	2	155	130.00	130/1 = 130
	3	145	142.50	(130+155)/2 = 142.50
Forecast of Demand in Period 5 (This forecast made after seeing Demand in Period 4)	4	160	143.33	(130+155+145)/3 = 143.33
	5	151	147.50	(130+155+145+160)/4 = 147.5
	6	143	148.20	(130+155+145+160+151)/5 = 148.2
	7		147.33	(130+155+145+160+151+143) / 6 = 147.33

Method 3: The Moving Average Forecast

$$F_t = (A_{t-n} + \dots + A_{t-2} + A_{t-1})/n$$

				Using n=3	
Period		Demand	Forecast		
Forecast of Demand in Period 2 (This forecast made after seeing Demand in Period 1)	1	130	-		
	2	155	-	Not enough history	
	3	145	-	Not enough history	
Forecast of Demand in Period 5 (This forecast made after seeing Demand in Period 4)	4	160	143.33	(130+155+145)/3 = 143.33	
	5	151	153.33	(155+145+160)/3 = 153.33	
	6	143	152.00	(145+160+151)/3 = 152.00	
	7		151.33	(160+151+143) /3 = 151.33	

Method 4: Weighted Moving Average Forecast

$$F_t = (w_{t-n+1} A_{t-n+1} + \dots + w_{t-2} A_{t-2} + w_{t-1} A_{t-1}) / \sum w$$

	Period	Demand	Forecast
Forecast of Demand in Period 2 (This forecast made after seeing Demand in Period 1)	1	130	-
	2	155	-
	3	145	-
Forecast of Demand in Period 5 (This forecast made after seeing Demand in Period 4)	4	160	145.00
	5	151	154.50
	6	143	152.50
	7		

n=3:
 $w_{t-1}=0.5$, $w_{t-2}=0.3$,
 $w_{t-3}=0.2$

Not enough history

Not enough history

$$0.2(130) + 0.3(155) + 0.5(145) = 145.00$$

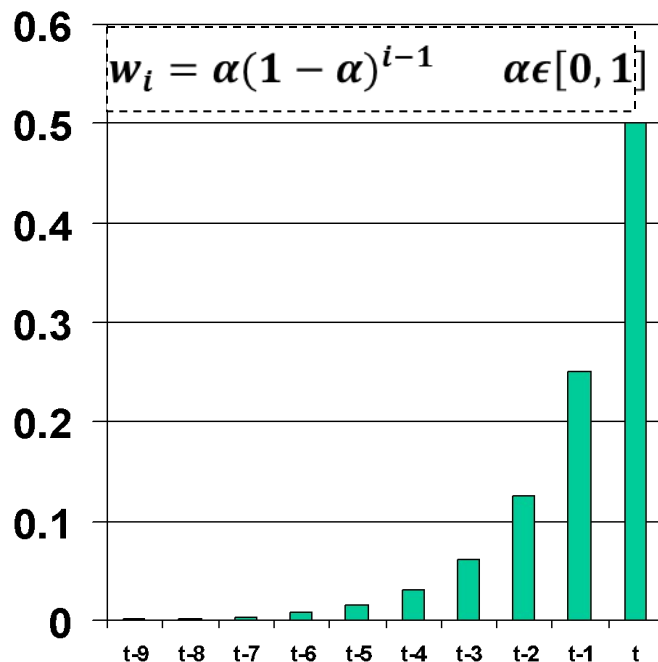
$$0.2(155) + 0.3(145) + 0.5(160) = 154.50$$

$$0.2(145) + 0.3(160) + 0.5(151) = 152.50$$

Method 5: Exponential Smoothing Forecast

Motivation

Consider a weighted moving average forecast with the following weights



Exponential smoothing is like a weighted average

$$F_t = \alpha A_{t-1} + (1 - \alpha) F_{t-1}$$

Is the same as

$$F_t = \sum_{i=1}^{t-1} w_i A_{t-i}$$

$$w_i = \alpha(1 - \alpha)^{i-1}$$

Method 5: Exponential Smoothing Forecast

$$F_t = \alpha A_{t-1} + (1 - \alpha)F_{t-1}$$

Period	Demand	Forecast
1	130	-
2		

$\alpha = .2$

Forecast of
Demand in
Period 2

(This forecast
made after
seeing Demand
in Period 1)

$$F_2 = \alpha A_1 + (1 - \alpha)F_1$$

$$F_2 = (0.2)130 + (1 - 0.2)F_1$$

Problem!

Didn't have a forecast for period 1 so how can we start the exponential smoothing method

Options

- (1) Use Naïve Forecast for Period 2 and then do exponential smoothing for periods 3,4,5,6,.....
- (2) Choose an initial forecast for period 1 (in some manner) and then do exponential smoothing for periods 2,3,4,5,6,.....

Exponential Smoothing Forecast: Using Naïve Method for Period 2

$\alpha = .2$

Period	Demand	Forecast
1	130	-
2	155	130
3	145	135.00
4	160	137.00
5	151	141.60
6	143	143.48
7		

Forecast of
Demand in
Period 2

(This forecast
made after
seeing Demand
in Period 1)

Forecast of
Demand in
Period 5

(This forecast
made after
seeing Demand
in Period 4)

Using Naive

$$0.2(155) + (1-0.2)(130) = 135.00$$

$$0.2(145) + (1-0.2)(135) = 137.00$$

$$0.2(160) + (1-0.2)(137) = 141.60$$

$$0.2(151) + (1-0.2)(141.60) = 143.48$$

Exponential Smoothing Forecast: Using Naïve Method for Period 2

$\alpha = .2$

Using Naive

Period	Demand	Forecast
1	130	-
2	155	130
3	145	135.00
4	160	137.00
5	151	141.60
6	143	143.48
7		143.38

Forecast of
Demand in
Period 2

(This forecast
made after
seeing Demand
in Period 1)

Forecast of
Demand in
Period 5

(This forecast
made after
seeing Demand
in Period 4)

$$0.2(155) + (1-0.2)(130) = 135.00$$

$$0.2(145) + (1-0.2)(135) = 137.00$$

$$0.2(160) + (1-0.2)(137) = 141.60$$

$$0.2(151) + (1-0.2)(141.60) = 143.48$$

$$0.2(143) + (1-0.2)(143.48) = 143.38$$

Exponential Smoothing Forecast: Forecast for Period 1 Is Given

$\alpha = .2$

Period	Demand	Forecast
1	130	140
2	155	138
3	145	141.40
4	160	141.12
5	151	145.70
6	143	146.76
7		146.00

Forecast of
Demand in
Period 2

(This forecast
made after
seeing Demand
in Period 1)

Forecast of
Demand in
Period 5

(This forecast
made after
seeing Demand
in Period 4)

$$0.2(130) + (1-0.2)(140) = 138.00$$

$$0.2(155) + (1-0.2)(138) = 141.40$$

$$0.2(145) + (1-0.2)(141.40) = 141.12$$

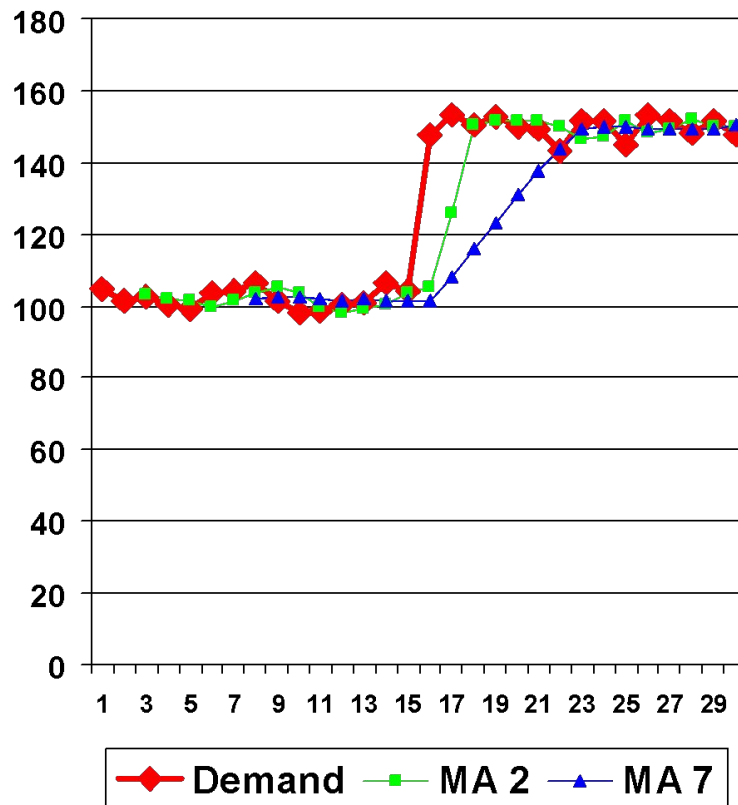
$$0.2(160) + (1-0.2)(141.12) = 145.70$$

$$0.2(151) + (1-0.2)(145.70) = 146.76$$

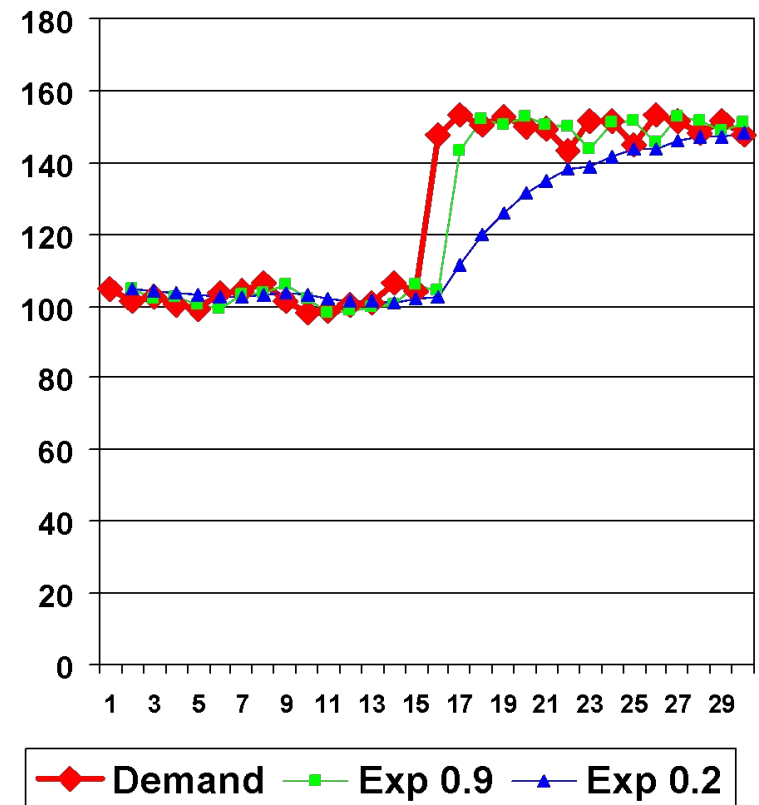
$$0.2(143) + (1-0.2)(146.76) = 146.00$$

Forecast Parameter Influences *Responsiveness* to Shift in Mean

Moving Average

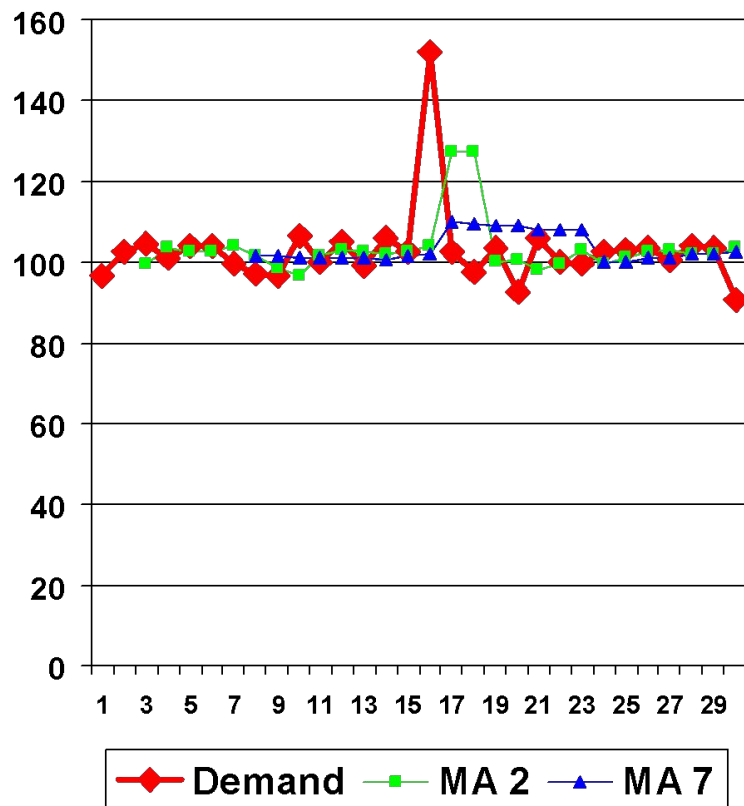


Exponential Smoothing

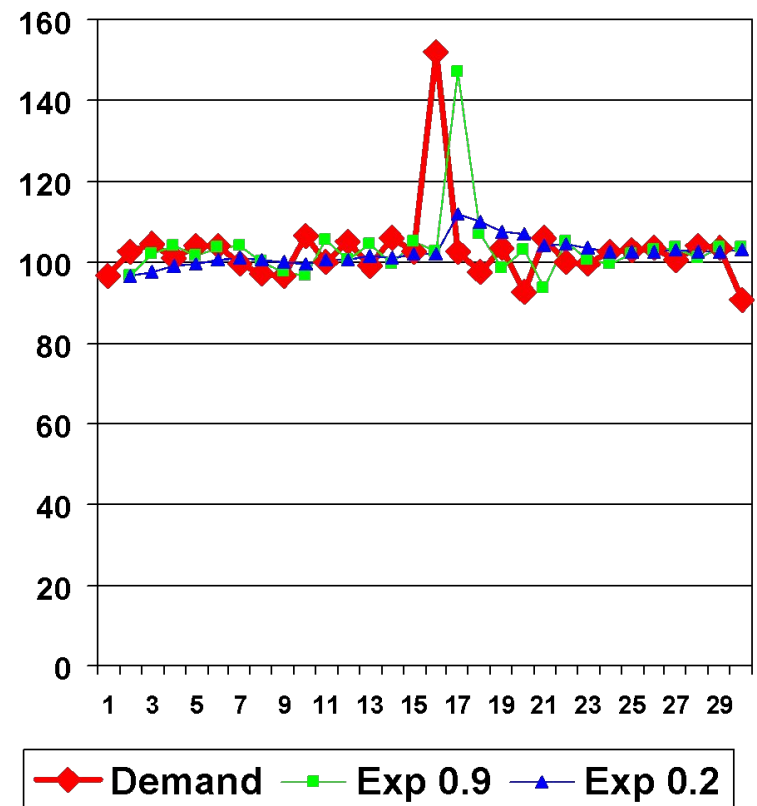


Forecast Parameter Influences *Stability* to Random Fluctuations

Moving Average

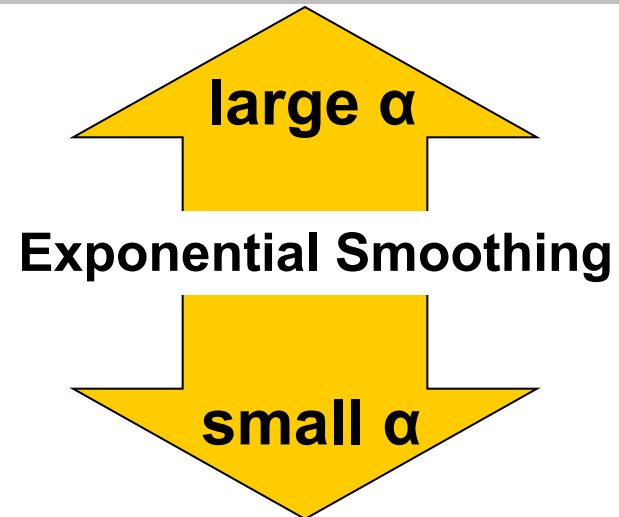
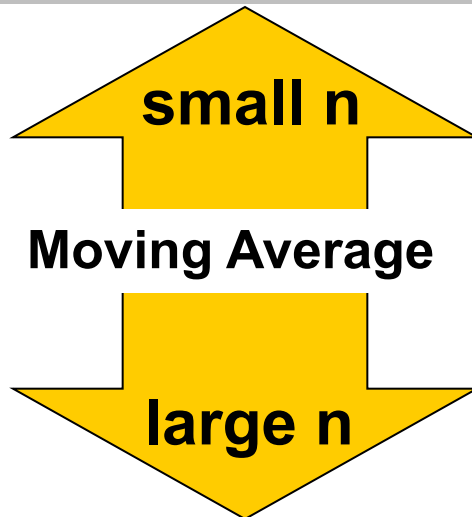


Exponential Smoothing



Responsiveness vs. Stability: Trade-off

Responsiveness: Ability of forecast to respond quickly to a true change in mean level demand

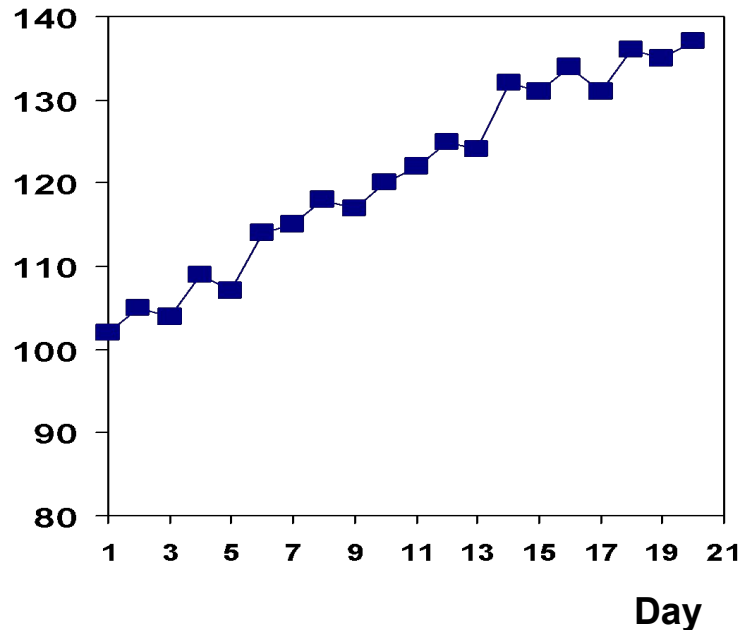


Stability: Ability of forecast to ignore random variations

Time Series With Trends

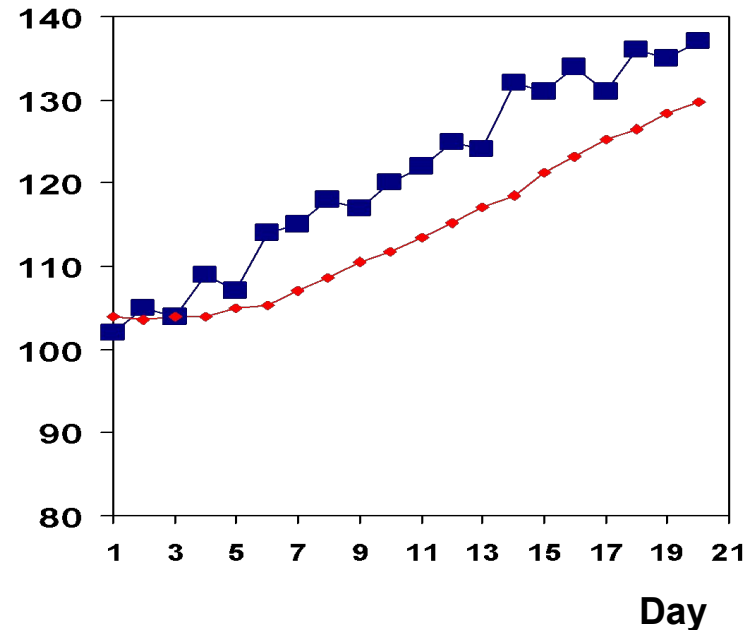
Consider demand in a growing market

Demand



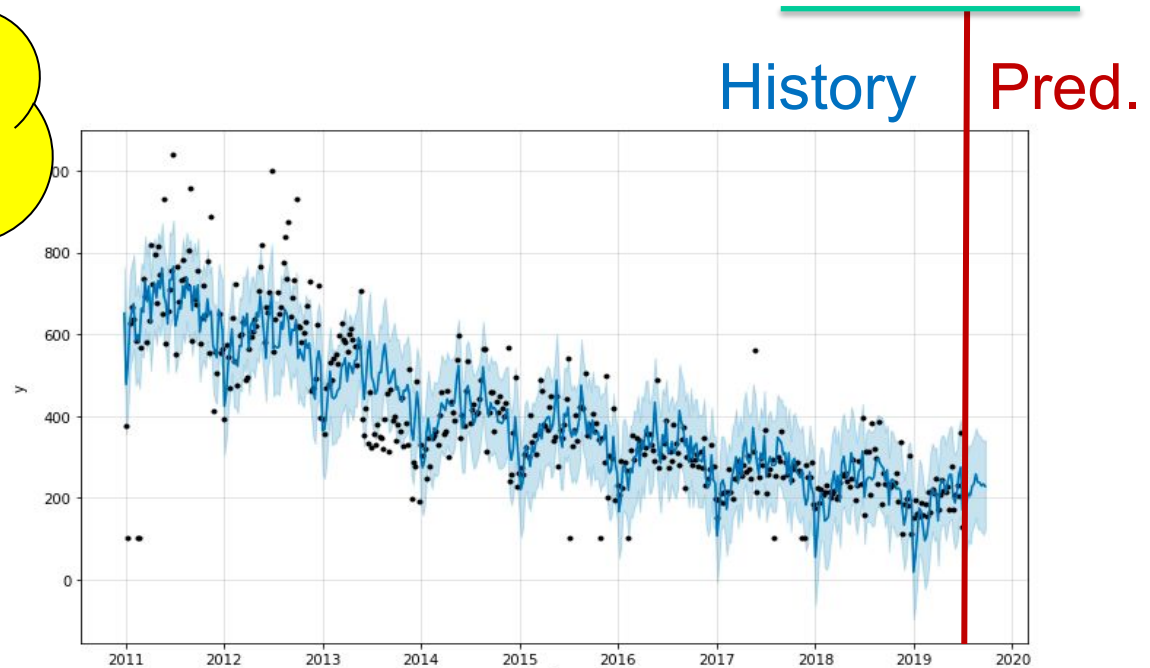
What would happen if we used the techniques from the last session ?

Demand



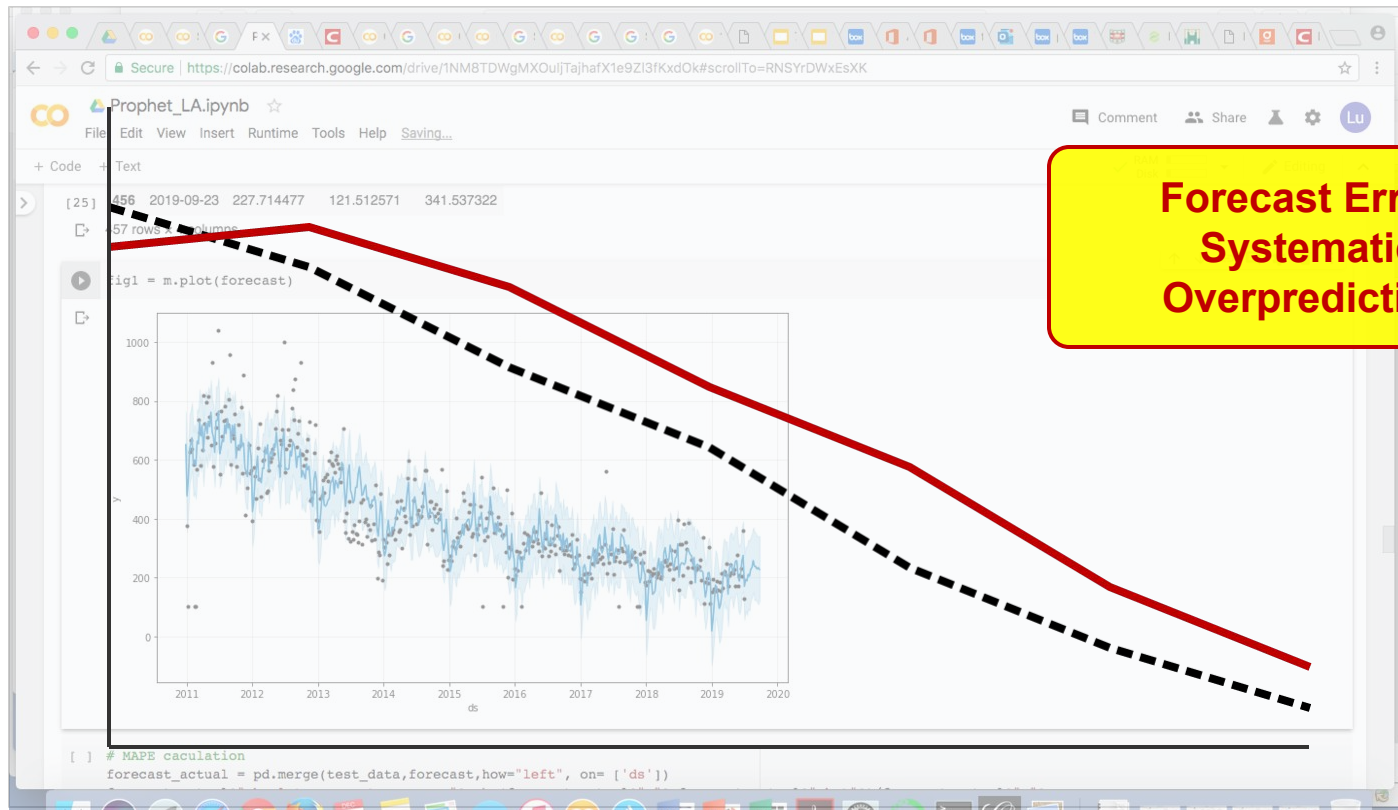
Demand in Declining Market

What would happen
if we applied
Exponential
Smoothing to the
A/B demand
pattern?



Simple Exponential Smoothing: Lag in Trends

In a declining market, simple exponential smoothing **overpredicts** demand because it lags behind the trend.



**Forecast Error:
Systematic
Overprediction**

 Actual Demand (Declining)

 Simple Exp. Smoothing (Lagging)

Exponential Smoothing with Trend Adjustment (Double Exponential Smoothing)

**When a trend is present,
exponential smoothing must be
modified**

MONTH	ACTUAL DEMAND	FORECAST (F_t) FOR MONTHS 1 – 5
1	100	$F_1 = 100$ (given)
2	200	$F_2 = F_1 + \alpha(A_1 - F_1) = 100 + .4(100 - 100) = 100$
3	300	$F_3 = F_2 + \alpha(A_2 - F_2) = 100 + .4(200 - 100) = 140$
4	400	$F_4 = F_3 + \alpha(A_3 - F_3) = 140 + .4(300 - 140) = 204$
5	500	$F_5 = F_4 + \alpha(A_4 - F_4) = 204 + .4(400 - 204) = 282$

Exponential Smoothing with Trend Adjustment

Forecast including trend $(F/T)_t =$ Exponentially smoothed forecast (F_t) + Exponentially smoothed trend (T_t)

$$F_t = \alpha(A_{t-1}) + (1 - \alpha)(F_{t-1} + T_{t-1})$$

$$T_t = \beta(F_t - F_{t-1}) + (1 - \beta)T_{t-1}$$

where F_t = exponentially smoothed forecast average

T_t = exponentially smoothed trend

A_t = actual demand

α = smoothing constant for average ($0 \leq \alpha \leq 1$)

β = smoothing constant for trend ($0 \leq \beta \leq 1$)

Exponential Smoothing with Trend Adjustment

Step 1: Compute F_t

Step 2: Compute T_t

Step 3: Calculate the forecast $FIT_t = F_t + T_t$

Exponential Smoothing with Trend Adjustment Example

MONTH (t)	ACTUAL DEMAND (A_t)	MONTH (t)	ACTUAL DEMAND (A_t)
1	12	6	21
2	17	7	31
3	20	8	28
4	19	9	36
5	24	10	?

$$\alpha = .2$$

$$\beta = .4$$

Exponential Smoothing with Trend Adjustment Example

TABLE 4.2

Forecast with $\alpha = .2$ and $\beta = .4$

MONTH	ACTUAL DEMAND	SMOOTHED FORECAST AVERAGE, F_t	SMOOTHED TREND, T_t	FORECAST INCLUDING TREND, FIT_t
1	12	11	2	13.00
2	17	12.80		
3	20			
4	19			
5	24			
6	21			
7	31			
8	28			
9	36			
10	—			

Step 1: Average for Month

$$\begin{aligned}
 F_2 &= \alpha A_1^2 + (1 - \alpha)(F_1 + T_1) \\
 F_2 &= (.2)(12) + (1 - .2)(11 + 2) \\
 &= 2.4 + (.8)(13) = 2.4 + 10.4 \\
 &= 12.8 \text{ units}
 \end{aligned}$$

Exponential Smoothing with Trend Adjustment Example

TABLE 4.2

Forecast with $\alpha = .2$ and $\beta = .4$

MONTH	ACTUAL DEMAND	SMOOTHED FORECAST AVERAGE, F_t	SMOOTHED TREND, T_t	FORECAST INCLUDING TREND, FIT_t
1	12	11	2	13.00
2	17	12.80	1.92	
3	20			
4	19			
5	24			
6	21			
7	31			
8	28			
9	36			
10	—			

Step 2: Trend for Month

$$T_2 = \beta(F_2 - F_1) + (1 - \beta)T_1$$

$$T_2 = (.4)(12.8 - 11) + (1 - .4)(2)$$

$$= .72 + 1.2 = 1.92 \text{ units}$$

Exponential Smoothing with Trend Adjustment Example

TABLE 4.2

Forecast with $\alpha = .2$ and $\beta = .4$

MONTH	ACTUAL DEMAND	SMOOTHED FORECAST AVERAGE, F_t	SMOOTHED TREND, T_t	FORECAST INCLUDING TREND, FIT_t
1	12	11	2	13.00
2	17	12.80	1.92	14.72
3	20			
4	19			
5	24			
6	21			
7	31			
8	28			
9	36			
10	—			

Step 3: Calculate FIT for Month

$$FIT_2 = F_2 + T_2$$

$$FIT_2 = 12.8 + 1.92$$

$$= 14.72 \text{ units}$$

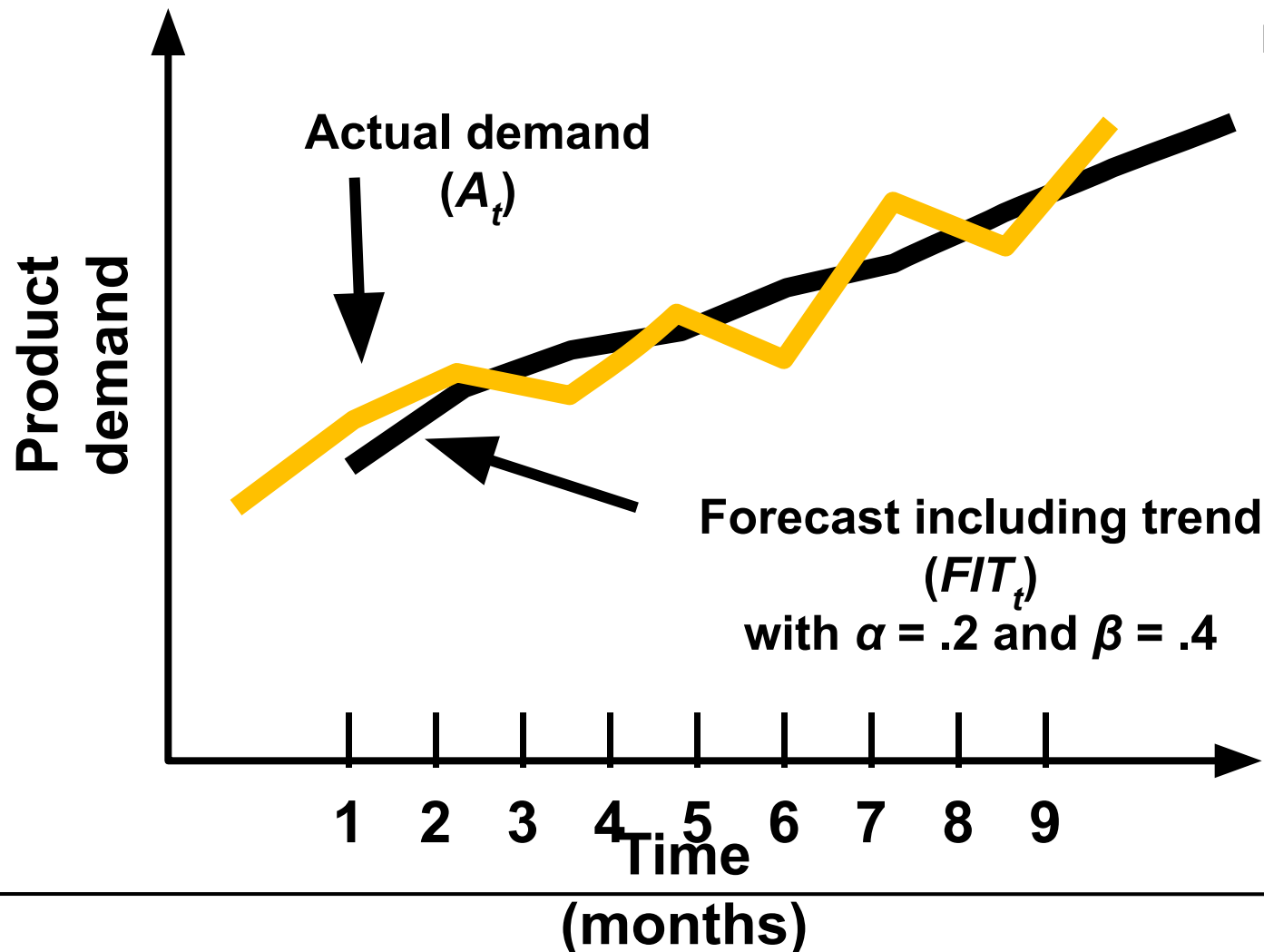
Exponential Smoothing with Trend Adjustment Example

TABLE 4.2

Forecast with $\alpha = .2$ and $\beta = .4$

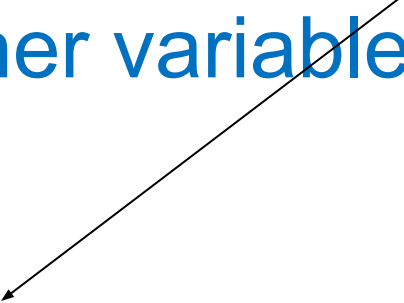
MONTH	ACTUAL DEMAND	SMOOTHED FORECAST AVERAGE, F_t	SMOOTHED TREND, T_t	FORECAST INCLUDING TREND, FIT_t
1	12	11	2	13.00
2	17	12.80	1.92	14.72
3	20	15.18	2.10	17.28
4	19	17.82	2.32	20.14
5	24	19.91	2.23	22.14
6	21	22.51	2.38	24.89
7	31	24.11	2.07	26.18
8	28	27.14	2.45	29.59
9	36	29.28	2.32	31.60
10	—	32.48	2.68	35.16

Exponential Smoothing with Trend Adjustment Example

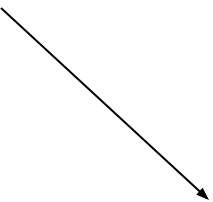


Causal Forecasting

- Assumption: **Variable we want to forecast** is related to **other variables in the environment**.



Dependent variable



Independent variable(s)

- In this course we consider models with a single independent variable
-

- Statistical method to model relationships between variables
- Predicts future outcomes and identifies key business drivers
- Simple linear: one predictor
- Multiple regression: multiple predictors
- Quantifies relationships for actionable forecasting

Key Market Factors for Forecasting

Variable Y depends on Variable X

- **Beach Attendance:**

Depends on the temperature.

- **Gasoline Price:**

Depends on world oil production.

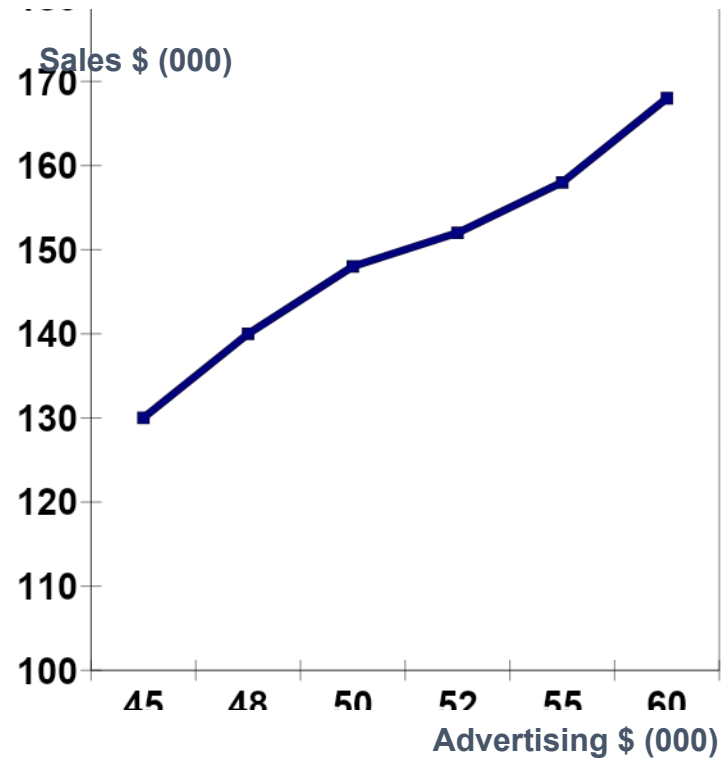
- **Sales Performance:**

Depends on advertising spend.

PRO TIP:

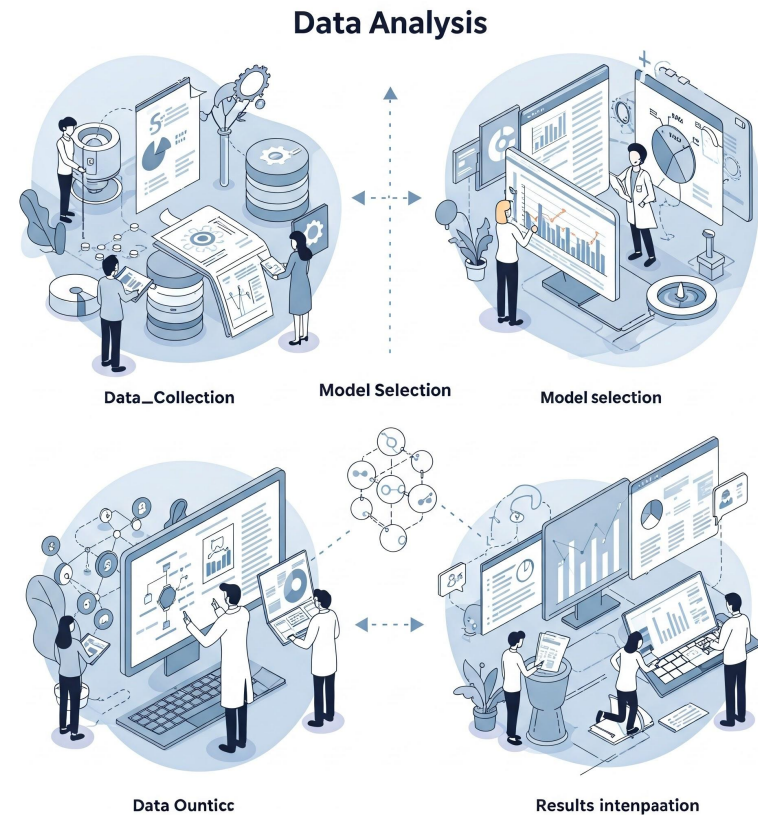
Recall the Advertising Budget from Class 1

The relationship might be linear



How Regression Works

- Collect Data: Gather data on dependent and independent variables.
- Choose Model: Select simple or multiple regression.
- Run Analysis: Use software (Excel, R, Python) to estimate.
- Interpret Results: Assess significant factors and impacts.
- Forecast: Apply model to predict future outcomes.



Linear Regression Output from Excel

Regression Statistics

Multiple R	0.9644
R Square	0.9300
Adjusted R Square	0.9125
Standard Error	3.9537
Observations	6.0000

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1.0000	830.8050	830.8050	53.1475	0.0019
Residual	4.0000	62.5283	15.6321		
Total	5.0000	893.3333			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	24.0660	17.2585	1.3944	0.2356	-23.8514	71.9834
Adverstising	2.4245	0.3326	7.2902	0.0019	1.5012	3.3479

Forecasting Demand: Coffee Shop

Scenario & Data

Predicting weekly demand based on ad spend and temperature.

Metrics: Cups sold, Ad spend (\$), Weekly temp (°F)

Regression Model

$$\text{Cups Sold} = 200 + 0.5 * \text{Ad Spend} - 2 * \text{Temp}$$

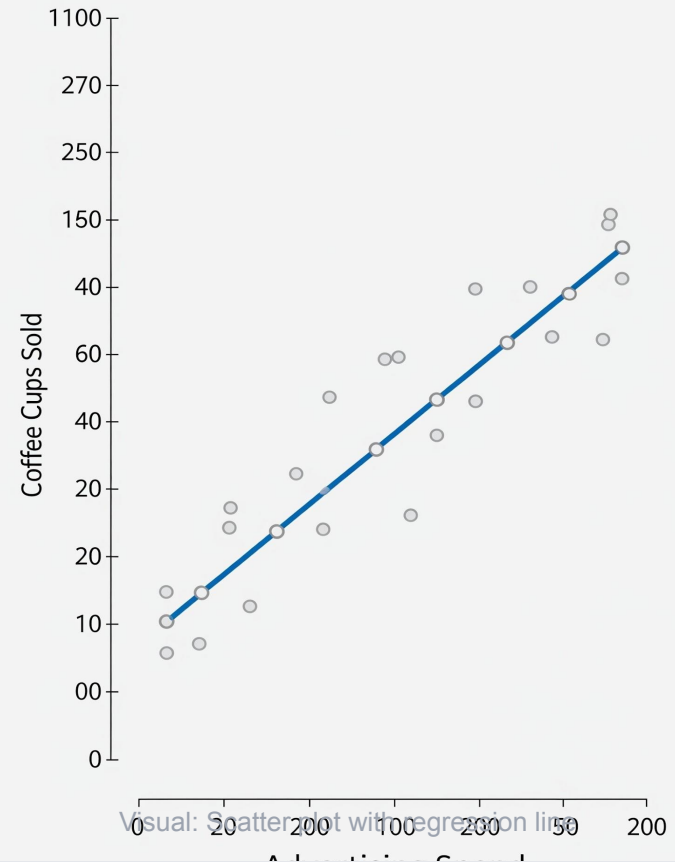
Key Interpretations

+\$1,000 Ad Spend

leads to **+500 cups sold**

+1°F Temperature

leads to **2 fewer cups sold**



Actionable Forecasting

Prediction

$\text{Cups Sold} = 200 + 0.5 * 2000 - 2 * 70 = \mathbf{1,060 \text{ cups}}$

Business Actions

-  Increase ad budget to boost demand
-  Adjust inventory for warmer weeks
-  Test scenarios (e.g., \$3,000 ad spend)

Validation



Compare predictions to actual data to refine the model.

Measuring Forecast Errors

Measuring Forecast Errors: Mean Forecast Error (MFE)

Period	Demand	Forecast	Error
1	130	-	-
2	155	130.00	25.00
3	145	155.00	-10.00
4	160	145.00	15.00
5	151	160.00	-9.00
6	143	151.00	-8.00
		Mean Forecast Error =	2.60

$$\begin{aligned} & (25 + (-10) + 15 + (-9) + (-8)) / 5 \\ & = \\ & 2.60 \end{aligned}$$

Measuring Forecast Errors: Mean Absolute Deviation (MAD)

Period	Demand	Forecast	Error	Absolute Error
1	130	-	-	-
2	155	130.00	25.00	25.00
3	145	155.00	-10.00	10.00
4	160	145.00	15.00	15.00
5	151	160.00	-9.00	9.00
6	143	151.00	-8.00	8.00
			MAD =	13.40

Measuring Forecast Errors:

Mean Absolute Percentage Error (MAPE)

Period	Demand	Forecast	Error	% Error	Abs % Error
1	130	-	-	-	-
2	155	130.00	25.00	$=25/155=16.13\%$	16.13%
3	145	155.00	-10.00	$=-10/145=-6.90\%$	6.9%
4	160	145.00	15.00	$=15/160=9.38\%$	9.38%
5	151	160.00	-9.00	$=-9/151=-5.96\%$	5.96%
6	143	151.00	-8.00	$=-8/143=-5.59\%$	5.59%
				MAPE	8.79%

Measuring Forecast Errors: Mean Squared Error (MSE)

Period	Demand	Forecast	Error	Squared Error
1	130	-	-	-
2	155	130.00	25.00	625.00
3	145	155.00	-10.00	100.00
4	160	145.00	15.00	225.00
5	151	160.00	-9.00	81.00
6	143	151.00	-8.00	64.00
			MSE =	219.00

MAD vs. MSE: Key Differences

Mean Absolute Deviation (MAD)

Treats all errors equally, regardless of magnitude.

Mean Squared Error (MSE)

Heavily penalizes large errors by squaring them.

Managerial Decision Scenario

If a manager prefers a method that generates small frequent errors over large infrequent errors...

Which approach should be used? MSE

Note: MSE is the superior choice here because its squared penalty discourages large "outlier" errors that managers want to avoid.

AI-Powered Forecasting: The Next Evolution

How Gemini Enhances Traditional Methods

- Traditional quantitative methods (Naïve, Moving Average, ES, Regression) are powerful but **manual**.
- Generative AI (Google Gemini) adds **speed, pattern detection**, and natural-language interaction.
- Gemini can analyze historical data, suggest optimal models (α , β), generate code, and run “**what-if**” **scenarios**.
- **Hybrid approach:** Use traditional models + AI assistance.

```
=FORECAST.ETS (...)  
=AVERAGE (B2 : B12)
```

Traditional Methods (Excel)



The result? More accurate forecasts in less time by combining human expertise with machine intelligence.

Using Gemini for Time Series Forecasting

1. Data Integration

Upload historical demand data (e.g., vaccine or detergent datasets).

2. Intelligent Analysis

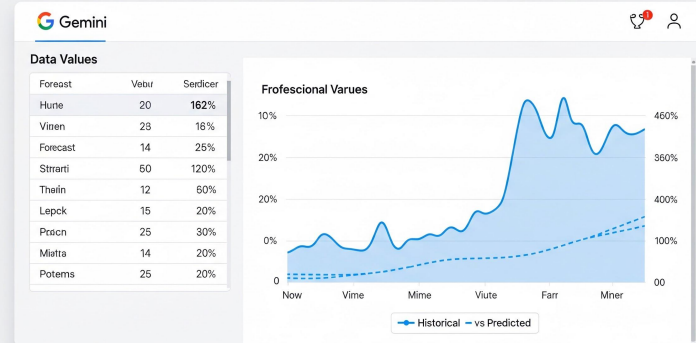
Returns instant forecasts, error metrics (MAD/MAPE/MSE), and charts.

Ideal for rapid sensitivity testing of α and β values.

AI-Powered Precision

Sample Prompts

- "Perform exponential smoothing ($\alpha=0.2$) and forecast periods 5-8"
- "Detect trend and suggest DES parameters"
- "Compare Naïve vs Moving Average on MAPE"



Gemini for Causal & Regression Forecasting

1. Multi-Variable Input

Feed in multiple variables (price, ad spend, temperature, income segment, etc.) to identify complex correlations.

2. Scenario Modeling

Combines perfectly with coffee-shop regression examples for real-time "What-If" analysis.

AI can generate Python code for advanced Monte Carlo simulations.

Predictive Power & Interpretability

Sample Causal Prompts

- "Run linear regression: Demand ~ Price + Ad Spend + Temperature. Interpret coefficients."
- "What happens to demand if we increase ad spend by \$2,000 and temperature rises 5°F?"
- "Identify key market drivers from this dataset and suggest optimal price point."

Gemini

Benefits, Limitations & Best Practices

Benefits



Extremely fast analysis of large/filtered datasets



Handles complex relationships and “what-if” scenarios instantly



Natural language → no coding required for students/managers

Limitations



Less transparent than simple statistical models



Can “hallucinate” – always validate with MAD/MAPE/MSE



Quality of output depends on quality of input data

Best Practice

Use Gemini as an assistant alongside your traditional methods (Naïve, ES, Regression)



Forecasting in Harvard's Data Analytics Simulation

Data-Driven Management of Blue Detergent

Managerial Context & Objectives

-

You manage **Blue detergent** for the Kelsey-White company in a competitive US marketplace.

-

Key Decision (Years 5–8): Determine the optimal number of units to produce each period.

-

The Goal: Maximize total operating profit by solving the classic **Newsvendor problem**.

-

Balance holding costs for excess inventory vs. opportunity costs of lost sales.

Forecast Demand Tool

Utilizes **Monte Carlo simulations** with triangular distributions based on regional estimates (min, expected, max).



How Your Course Forecasting Models Help Set Better Inputs

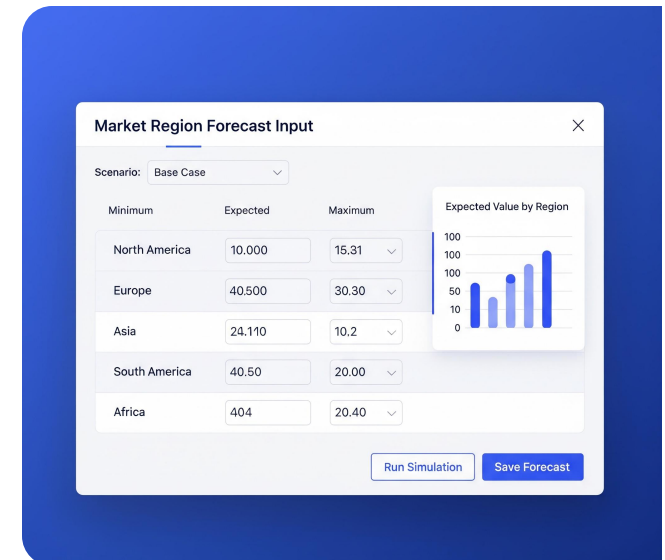
Bridging Theory to Simulation

Time-series methods (Moving Average, Exponential Smoothing, Double ES): Use Years 1–4 historical demand to estimate “Expected” demand and detect trends.

Causal/Regression analysis: Adjust forecasts using income, age, region, price, media spend, formulation (pods), and positioning (odor elimination).

Error metrics (MAD/MAPE/MSE): Test and refine your triangular distribution parameters (min/max).

Result: Far more accurate min/expected/max values than guessing.



Simulation Forecast Tool

Enhanced Decision Support: Turning Classroom Models into Profitable Simulation Actions

Balancing Forecasts Against Inventory Costs

Optimizing the Newsvendor Trade-off: Overage vs. Underage

Cost Dynamics & The Critical Ratio

Cost of Underage (Cu): The cost of being "too low"

- Lost profit contribution: Price - Cost
- Impact: Opportunity cost of lost sales

Cost of Overage (Co): The cost of being "too high"

- Loss per unit: Cost - Salvage Value
- Impact: Holding costs & liquidation

Optimal Service Level (Critical Ratio)

$$CR = Cu / (Cu + Co)$$



Key Decision Objective

Maximize total operating profit by finding the "Sweet Spot" where incremental cost of underage equals incremental cost of overage.

Forecasting Integration

Use Time-Series and Regression models to build the demand distribution (μ , σ) then apply CR to set the order quantity.

Strategic Insight: The optimal quantity is usually higher than the mean forecast when profit margins are high.

Better Forecasting = Higher Simulation Scores

Turning Predictive Accuracy into Competitive Advantage

The Profit Engine



Accurate Forecasts: Drives optimal production quantity, balancing the Newsvendor trade-off.



Cost Efficiency: Minimizes the double-hit of excess inventory and missed sales (stockouts).



Performance: Directly impacts cumulative profit, revenue growth, and market share.

Top-Tier Student Strategy:

- **Descriptive:** Data Explorer analysis.
- **Predictive:** Try to utilize course models.
- **Prescriptive:** Profit-maximizing production.

Simulation Benchmarks



Takeaway: The forecasting tools you learned today are exactly what top scorers use in the Harvard simulation!